

Fig. 6: An actual-size, single-side PCB for the microcontroller-based intelligent traffic light system

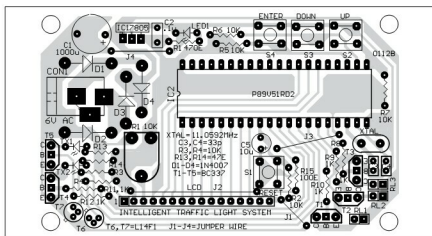


Fig. 7: Component layout for the PCB

and assembled using A51 assembler. The generated hex code is burnt into the microcontroller using a suitable

programmer. Flash Magic programmer can also be used to burn the hex code through a serial cable. The program works as per the flow-chart shown in Fig. 5.

The default values of system parameters like traffic profile, vehicle count and traffic delays for respective profile are stored at memory location 2000H. The two-digit controlling interval is stored at memory location 2080H. Storing the traffic vehi-

cle count/traffic delays and controlling interval in two separate blocks allows separate updation of the required parameter. Whenever the configuration parameters are changed through switches and LCD interaction, different memory blocks are erased and updated parameters written to the same.

Sub-routine 'erase_config' erases memory blocks 2000H through 207FH, which are used for storing the traffic profiles, vehicle counts and traffic delays for respective profiles. Sub-routine 'erase_cntrlint' erases memory blocks 2080H through 20FFH, which are used for storing the controlling interval. Sub-routine 'write_flash' handles writing any byte stored in the accumulator to any memory location.

Timer T0 of the microcontroller is configured to roll-over at a delay of 10 milliseconds. The roll-over is handled by 'intr_service' interrupt service routine. One-second delay formed by 10-milliseconds delay of timer T0 is used to manage the functionality of the entire system.

Construction and testing

An actual-size, single-side PCB for the microcontroller-based intelligent traffic light system is shown in Fig. 6 and its component layout in Fig. 7. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use proper IC base for the microcontroller. Before inserting the ICs, check the supply voltage. Align infrared LEDs and photo-transistors such that these directly face each other. Cover the photo-transistor with a mask to protect it from light. Using preset VR1, set the LCD contrast for proper display.

EFY note. The source code of this article is available on www.efymag.com website. ●

TABLE I
Default Parameter Values

Parameter	Values
Vehicle count for low profile	05 vehicles
Vehicle count for medium profile	10 vehicles
Vehicle count for high profile	15 vehicles
Red light delay for low profile	011 seconds
Yellow light delay for low profile	006 seconds
Green light delay for low profile	009 seconds
Red light delay for medium profile	010 seconds
Yellow light delay for medium profile	006 seconds
Green light delay for medium profile	011 seconds
Red light delay for high profile	008 seconds
Yellow light delay for high profile	005 seconds
Green light delay for high profile	013 seconds
Control interval	01 minute

TABLE II
Parameter Ranges

Parameter	Range
Vehicle count	00 to 99 vehicles
Traffic lights delays	001 to 999 seconds
Control interval	01 to 99 minutes

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